

Unit-2 MongoDB PPT.pptx

Exported: May 28, 2026

Summary

This educational material serves as an introductory guide to **MongoDB**, a prominent **NoSQL database** designed for managing big data and real-time web applications. The text contrasts traditional relational databases with MongoDB's **document-oriented architecture**, highlighting its use of **BSON** and a **flexible schema** that allows for horizontal scaling through **sharding and replication**. A significant portion of the content explores the **CAP Theorem**, explaining the necessary trade-offs between **consistency, availability, and partition tolerance** in distributed systems. Finally, the source provides practical instruction on performing **CRUD operations**, managing collections, and utilizing specialized features like **GridFS** for binary storage.

Keywords: MongoDB Introduction, NoSQL Database Types, CAP Theorem, CRUD Operations, Sharding and Replication

Content

Big DataAnd AnalyticsBig Data and Analytics by Seema Acharya and Subhashini ChellappanCopyright 2015, WILEY INDIA PVT. LTD.Seema AcharyaSubhashini ChellappanChapter 6Big Data and Analytics by Seema Acharya and Subhashini ChellappanCopyright 2015, WILEY INDIA PVT. LTD.Introduction to MongoDBLearning Objectives and Learning OutcomesBig Data and Analytics by Seema Acharya and Subhashini ChellappanCopyright 2015, WILEY INDIA PVT. LTD.Learning ObjectivesLearning OutcomesIntroduction to MongoDBTo study the features of MongoDB.To learn how to perform CRUD operations.To study aggregation.To study the MapReduce Framework.To import from and export to CSV format.To comprehend the reasons behind the popularity of NoSQL database.To be able to perform CRUDOperations.To comprehend MapReduce framework.To understand the aggregation.To be able to successfully import from and export to CSV.Session PlanBig Data and Analytics by Seema Acharya and Subhashini ChellappanCopyright 2015, WILEY INDIA PVT. LTD.Lecture time90 to 120 minutesQ/A15 minutesAgendaBig

Data and Analytics by Seema Acharya and Subhashini Chellappan Copyright 2015, WILEY INDIA PVT. LTD. NoSQL: Types of NoSQL databases CAP Theorem Terms used in RDBMS and MongoDB MongoDB Query Language: CRUD Finding documents based on search criteria Dealing with Null values Count, Limit, Sort Skip, Aggregate Functions MongoDB- An Introduction Big Data and Analytics by Seema Acharya and Subhashini Chellappan Copyright 2015, WILEY INDIA PVT. LTD. Types of NoSQL Key value datastore Riak Redis Membase Column-oriented data store Cassandra HBase HyperTable Document datastore MongoDB CouchDB RavenDB Graph datastore InfiniteGraph Neo4 Allegro Graph Big Data and Analytics by Seema Acharya and Subhashini Chellappan Copyright 2015, WILEY INDIA PVT. LTD. { _id: ObjectId("5effaa5662679b5af2c58829"), email: "email@example.com", name: { given: "Jesse", family: "Xiao"}, age: 31, addresses: [{ label: "home", street: "101 Elm Street", city: "Springfield", state: "CA", zip: "90000", country: "US"}, { label: "mom", street: "555 Main Street", city: "Jonestown", province: "Ontario", country: "CA"}] } Brewer's CAP Theorem CAP Consistency Availability Partition tolerance Consistency implies that every read fetches the last write. Availability implies that reads and writes always succeed. In other words, each non-failing node will return a response in a reasonable amount of time. Partition tolerance implies that the system will continue to function when network partition occurs. When to choose consistency over availability and vice-versa 1. Choose availability over consistency when your business requirements allow some flexibility around when the data in the system synchronizes. 2. Choose consistency over availability when your business requirements demand atomic reads and writes. CAP Theorem CAP Theorem CAP Theorem CAP Theorem CAP Theorem CAP Theorem If the bank prioritizes consistency, ATM may refuse to process deposits or withdrawals until the Partition is resolved. This ensures that the balance remains consistent, but the system is unavailable to the Customer. CAP Theorem If the Bank prioritize Availability, the ATM may allow withdrawals and deposits to occur, but the data Remain Inconsistent until the Partition Tolerance is resolved. CAP theorem When there is a network Partition, The Customer could withdraw the entire balance from both the ATMs. CAP theorem When the Network comes back online, the inconsistency is resolved and now the balance is negative. That is not good. What is MongoDB? Big Data and Analytics by Seema Acharya and Subhashini Chellappan Copyright 2015, WILEY INDIA PVT. LTD. MongoDB is: Cross-platform- .Open source. Non-relational. Distributed. NoSQL. Document-oriented data store. Why MongoDB? Big Data and Analytics by Seema Acharya and Subhashini Chellappan Copyright 2015, WILEY INDIA PVT. LTD. Why MongoDB? Big Data and Analytics by Seema Acharya and Subhashini Chellappan Copyright 2015, WILEY INDIA PVT. LTD. Open Source Distributed- storing data across multiple servers or computers instead of just one Fast In-Place Updates- A method of modifying data within a database or data

structure where only the specific parts that need changing are altered directly, without needing to read the entire data set or create a temporary copy

Replication- the process of copying data from a primary server to multiple secondary servers

Full Index Support -MongoDB supports many types of indexes, including single-field, compound, multikey, geospatial, text search, hashed, and wildcard indexes. These indexes help MongoDB execute queries efficiently.

Rich Query Language - a query language that supports a wide range of queries and can search and retrieve data based on specific conditions.

Easy Scalability- Easy scalability primarily through its built-in support for “horizontal scaling” using a feature called “sharding,” which allows you to distribute data across multiple servers by splitting it into smaller chunks,

Auto sharding- MongoDB automatically distributes data across multiple shards, allowing the database system to handle large datasets and high user demands

JSONBig Data and Analytics by Seema Acharya and Subhashini ChellappanCopyright 2015, WILEY INDIA PVT. LTD.Using JSONJSON, or JavaScript Object Notation, is a human-readable data interchange format.JSON objects are associative containers, wherein a string key is mapped to a value (which can be a number, string, boolean, array, an empty value — null, or even another object).

BSON, or Binary JSON, is the data format that MongoDB uses to organize and store data.

Creating or Generating a Unique KeyEach JSON document should have a unique identifier.It is similar to the primary key of relational databases.This facilitates search for documents based on the unique identifier.

ObjectIds use 12 bytes of storage, which gives them a string representation that is 24 hexadecimal digits: 2 digits for each byte.

Support for Dynamic QueriesBig Data and Analytics by Seema Acharya and Subhashini ChellappanCopyright 2015, WILEY INDIA PVT. LTD.MongoDB has extensive support for dynamic queries.This is in keeping with traditional RDBMS wherein we have static data and dynamic queries.”Static data” refers to a fixed dataset that doesn’t change frequently”dynamic queries” are queries that can be constructed and executed at runtime

Example: Static data:A database table containing a list of all countries with their corresponding country codes.

Dynamic query:A query that retrieves a list of customers from this table based on a user-selected region, where the region filter is provided dynamically when the user makes a selection.

Storing Binary DataBig Data and Analytics by Seema Acharya and Subhashini ChellappanCopyright 2015, WILEY INDIA PVT. LTD.MongoDB provides GridFS to support the storage of binary data. It can store up to 4 MB of data. This suffices for photographs or small video clips. If one wishes to store movie clips, it stores the metadata in a collection called “file”. It then breaks the data into small pieces called chunks and stores it in the chunks collection.

Replication in MongoDBClient

ApplicationPrimarySecondarySecondarySecondaryReplicationReplicationWritesReadsReplication

Big Data and Analytics by Seema Acharya and Subhashini ChellappanCopyright 2015, WILEY INDIA PVT. LTD.MongoDB provides data redundancy and high availability.It

helps to recover from hardware failure and service interruption. Sharding in MongoDB (256 GB) (256 GB) Logical Database (Collection 1) Collection 11 TB database Shard 1 (256 GB) Shard 2 Shard 3 Shard 4 (256 GB) Big Data and Analytics by Seema Acharya and Subhashini Chellappan Copyright 2015, WILEY INDIA PVT. LTD. (256 GB) (256 GB) Sharding is a method for distributing or partitioning data across multiple machines. Horizontal scaling, also known as scale-out, refers to adding machines to share the data set and load. Sharding in MongoDB In MongoDB, a sharded cluster consists of: Shards Mongos Config servers A shard is a replica set that contains a subset of the cluster's data. The mongos acts as a query router for client applications, handling both read and write operations. It dispatches client requests to the relevant shards and aggregates the result from shards into a consistent client response. Clients connect to a mongos, not to individual shards. Config servers are the authoritative source of sharding metadata. The sharding metadata reflects the state and organization of the sharded data. The metadata contains the list of sharded collections, routing information, etc. Advantages of Sharding Increased read/write throughput. Increased storage capacity. Data Locality. Terms Used in RDBMS and MongoDB Big Data and Analytics by Seema Acharya and Subhashini Chellappan Copyright 2015, WILEY INDIA PVT. LTD. Terms Used in RDBMS and MongoDB Big Data and Analytics by Seema Acharya and Subhashini Chellappan Copyright 2015, WILEY INDIA PVT. LTD. RDBMS MongoDB Database Database Table Collection Record Document Columns Fields / Key Value pairs Index Index Joins Embedded documents Primary Key Primary key (_id is a identifier) Data Types in MongoDB Big Data and Analytics by Seema Acharya and Subhashini Chellappan Copyright 2015, WILEY INDIA PVT. LTD. Data Types in MongoDB Big Data and Analytics by Seema Acharya and Subhashini Chellappan Copyright 2015, WILEY INDIA PVT. LTD. String Must be UTF-8 valid. Most commonly used data type. Integer Can be 32-bit or 64-bit (depends on the server). Boolean To store a true/false value. Double To store floating point (real values). Min/Max keys To compare a value against the lowest or highest BSON elements. Arrays To store arrays or list or multiple values into one key. Timestamp To record when a document has been modified or added. Null To store a NULL value. A NULL is a missing or unknown value. Date To store the current date or time in Unix time format. One can create object of date and pass day, month and year to it. Object ID To store the document's id. Binary data To store binary data (images, binaries, etc.). Code To store javascript code into the document. Regular expression To store regular expression. CRUD in MongoDB Big Data and Analytics by Seema Acharya and Subhashini Chellappan Copyright 2015, WILEY INDIA PVT. LTD. Collections Big Data and Analytics by Seema Acharya and Subhashini Chellappan Copyright 2015, WILEY INDIA PVT. LTD. To create a collection by the name "Person". Let us take a look at the collection list prior to the creation of the new collection "Person". `db.createCollection("Person");` Collections Big Data and Analytics by Seema

Acharya and Subhashini Chellappan Copyright 2015, WILEY INDIA PVT. LTD. To drop a collection by the name "student".db.student.drop(); To drop a database db.dropDatabase() Update Method Big Data and Analytics by Seema Acharya and Subhashini Chellappan Copyright 2015, WILEY INDIA PVT. LTD. Insert the document for "Aryan David" into the Students collection only if it does not already exist in the collection. However, if it is already present in the collection, then update the document with new values. (Update his Hobbies from "Skating" to "Chess".) Use "Update else insert" (if there is an existing document, it will attempt to update it, if there is no existing document then it will insert it).db.Students.update({_id:3, StudName:"Aryan David", Grade: "VII"},{\$set:{Hobbies: "Skating"}},{upsert:true}); Find Method Big Data and Analytics by Seema Acharya and Subhashini Chellappan Copyright 2015, WILEY INDIA PVT. LTD. To search for documents from the "Students" collection based on certain search criteria.db.Students.find({StudName:"Aryan David"}); Find Method Big Data and Analytics by Seema Acharya and Subhashini Chellappan Copyright 2015, WILEY INDIA PVT. LTD. To display only the StudName and Grade from all the documents of the Students collection. The identifier _id should be suppressed and NOT displayed.db.Students.find({}, {StudName:1, Grade:1, _id:0}); To display the StudName, Grade as well the identifier, _id from the document of the Students collection where the _id column is 1.db.Students.find({_id:1}, {StudName:1, Grade:1}); To display the StudName, Grade from the document of the Students collection where the _id column is 1. The _id field should not be displayed.db.Students.find({_id:1}, {StudName:1, Grade:1, _id:0}); Find Method Big Data and Analytics by Seema Acharya and Subhashini Chellappan Copyright 2015, WILEY INDIA PVT. LTD. To find those documents where the Grade is set to 'VII'.db.Students.find({Grade:{\$eq:'VII'}}).pretty(); To find those documents where the Grade is not set to 'VII'.db.Students.find({Grade:{\$ne:'VII'}}).pretty(); Find Method Big Data and Analytics by Seema Acharya and Subhashini Chellappan Copyright 2015, WILEY INDIA PVT. LTD. To find those documents from the Students collection where the Hobbies is set to either 'Chess' or is set to 'Skating'.db.Students.find ({Hobbies :{ \$in: ['Chess','Skating']}}).pretty (); To find those documents from the Students collection where the Hobbies is set neither to 'Chess' nor is set to 'Skating'.db.Students.find ({Hobbies :{ \$nin: ['Chess','Skating']}}).pretty (); Find Method To find those documents from the Students collection where the Hobbies is set to 'Graffiti' and the StudName is set to 'Hersch Gibbs' (AND condition).db.Students.find ({Hobbies :'Graffiti', StudName:'Hersch Gibbs'}).pretty(); Find Method Big Data and Analytics by Seema Acharya and Subhashini Chellappan Copyright 2015, WILEY INDIA PVT. LTD. To find documents from the Students collection where the StudName begins with "M".db.Students.find({StudName:/^M/}).pretty(); OR db.Students.find({StudName: {\$regex:"^M"}}).pretty(); To find documents from the Students collection where the

StudNameEnds in "s".db.Students.find({StudName:/s\$/}).pretty();Find MethodBig Data and Analytics by Seema Acharya and Subhashini ChellappanCopyright 2015, WILEY INDIA PVT. LTD.To find documents from the Students collection where the StudName has an "e" in any position.db.Students.find({StudName:/e/}).pretty();OR db.Students.find({StudName:/.*e.*\$/}).pretty();ORdb.Students.find({StudName: {\$regex:"e"}}).pretty();Find methodTo find those documents from the Students collection where the StudName ends in "a".db.Students.find({StudName: {regex:"a"}}).pretty();Dealing with NULL valuesTo add a new field with null value in existing documents(_id:3 and _id:4) of Students collection. A NULL is a missing or unknown value. When we place NULL as a value for a specified field , it implies that currently we do not know the value or the value is missing. We can always update the value of the field once we know it.Before we execute the commands to update documents with a null value in a column, let us first view the two documents. db.Students.find({\$or:[{_id:3},{_id:4}]});Update the documents with NULL values in the "Location" column.db.Students.update({_id:3},{ \$set: {Location:null}});db.Students.update({_id:4},{ \$set: {Location:null}});Dealing with NULL valuesTo search for NULL values in the Location column.db.Students.find({Location:{\$eq:null}});To remove "Location" field having "NULL" values from the documents (_id:3 and _id:4) from the Students collection.db.Students.update({_id:3},{ \$unset: {Location:null}});db.Students.update({_id:4},{ \$unset: {Location:null}});Count methodBig Data and Analytics by Seema Acharya and Subhashini ChellappanCopyright 2015, WILEY INDIA PVT. LTD.To find the number of documents in the Students collection.db.Students.count();To find the number of documents in the Students collection wherein the Grade is VII .db.Students.count({Grade:"VII"});Sort MethodBig Data and Analytics by Seema Acharya and Subhashini ChellappanCopyright 2015, WILEY INDIA PVT. LTD.To sort the documents from the Students collection in the ascending order of StudName.db.Students.find().sort({StudName:1}).pretty();To sort the documents from the Students collection in the descending order of StudName.db.Students.find().sort({StudName:-1}).pretty();db.Students.find().sort({Grade: 1,Hobbies:-1}).pretty();To sort the documents from the Students collection first on Grade in ascending order and then on Hobbies in descending order.db.Students.find().sort({Grade:1,Hobbies:1}).pretty();To sort the documents from the Students collection first on Grade in ascending order and then on Hobbies in ascending order.Skip MethodTo skip the first 2 documents from the Students collection.db.Students.find().skip(2).pretty();To sort the documents from the Students collection and skip the first document from the output.db.Students.find().skip(1).pretty().sort({StudName:1});To display the last 2 documents from the Students collection.db.Students.find().pretty().skip(db.Students.count()-2);To retrieve the

third, fourth and fifth document from the Students collection.
`collection.db.Students.find().pretty().skip(2).limit(3);`
Adding new field – update method
Big Data and Analytics by Seema Acharya and Subhashini Chellappan
 Copyright 2015, WILEY INDIA PVT. LTD. To add a new field to an existing document in MongoDB, you can use the `updateOne()` or `updateMany()` method along with the `$set` operator.
`db.collection.updateOne( // Filter to select the document(s) you want to update { <filter> }, // Update operation { // $set operator to add a new field $set: { newField: "value" // Specify the new field and its value } });`
`db.users.updateOne( { name: "John" }, // Filter documents where name is "John" { $set: { age: 30 } } // Add the new field "age" with value 30);`
Removing an existing field
Big Data and Analytics by Seema Acharya and Subhashini Chellappan
 Copyright 2015, WILEY INDIA PVT. LTD. To remove an existing field from an existing document in MongoDB, you can use the `updateOne()` or `updateMany()` method along with the `$unset` operator.
`db.collection.updateOne( // Filter to select the document(s) you want to update { <filter> }, // Update operation { // $unset operator to remove a field $unset: { fieldName: "" // Specify the name of the field you want to remove } });`
`db.users.updateOne( { name: "John" }, // Filter documents where name is "John" { $unset: { age: "" } } // Remove the "age" field);`
Save Method
Big Data and Analytics by Seema Acharya and Subhashini Chellappan
 Copyright 2015, WILEY INDIA PVT. LTD. `Save()` method will insert a new document if the document with the specified `_id` does not exist. However, if a document with specified `id` exists, it replaces the existing document with the new one.
`// Update an existing document with _id "123abc"`
`db.collection.save({ _id: "123abc", name: "Updated Name", age: 30, status: "active"});`
`// Insert a new document if _id "456def" doesn't already exist`
`db.collection.save({ _id: "456def", name: "New Document", age: 25, status: "inactive"});`
Remove Method
Big Data and Analytics by Seema Acharya and Subhashini Chellappan
 Copyright 2015, WILEY INDIA PVT. LTD. `Remove()` method is used to delete documents from a collection that match a specified query. It can remove either a single document or multiple documents depending on the criteria provided.
`db.collection.remove( <query>, { justOne: <boolean>, writeConcern: <document> } )`
`db.users.remove( { name: "John" } );`
`db.users.remove( { name: "John" }, { justOne: true } );`
Aggregate Function
Big Data and Analytics by Seema Acharya and Subhashini Chellappan
 Copyright 2015, WILEY INDIA PVT. LTD. `Aggregate Function`
`{_id: "C123", TotAccBal: 1400} { _id: "C111", TotAccBal: 1200 }`
Big Data and Analytics by Seema Acharya and Subhashini Chellappan
 Copyright 2015, WILEY INDIA PVT. LTD. `Customers`
`{ CustID: "C123", AccBal: 500, AccType: "S" }`
`{ CustID: "C123", AccBal: 900, AccType: "S" }`
`{ CustID: "C111", $match: { AccBal: 1200, AccType: "S" } }`
`{ CustID: "C123", AccBal: 1500, AccType: "C" }`
`{ CustID: "C123", AccBal: 500, AccType: "S" }`
`{ CustID: "C123", AccBal: 900, AccType: "S" }`
`$group { } { CustID: "C111", AccBal: 1200, AccType: "S" }`
Aggregate Function
Big Data and Analytics by

Seema Acharya and Subhashini ChellappanCopyright 2015, WILEY INDIA PVT. LTD.To group on “CustID” and compute the sum of “AccBal”, use the below syntax:`db.Customers.aggregate({$group:{_id: “$CustID”,“TotAccBal: {$sum:”$AccBal”}}});`First filter on “AccType:S” and then group it on “CustID” and then compute the sum of “AccBal” and then filter those documents wherein the “TotAccBal” is greater than 1200, use the below syntax:`db.Customers.aggregate( { $match : {AccType : “S” } },{ $group : { _id : “$CustID”,TotAccBal : { $sum : “$AccBal” } } },{ $match : {TotAccBal : { $gt : 1200 } } });`Aggregate FunctionBig Data and Analytics by Seema Acharya and Subhashini ChellappanCopyright 2015, WILEY INDIA PVT. LTD.To group on “CustID” and compute the average of the “AccBal” for each group:`db.Customers.aggregate({ $group : { _id : “$CustID”,TotAccBal : { $avg : “$AccBal” } } });`To group on “CustID” and determine the maximum “AccBal” for each group:`db.Customers.aggregate({ $group : { _id : “$CustID”,TotAccBal : { $max : “$AccBal” } } });`To group on “CustID” and determine the minimum “AccBal” for each group:`db.Customers.aggregate({ $group : { _id : “$CustID”,TotAccBal : { $min : “$AccBal” } } });`MongoImportBig Data and Analytics by Seema Acharya and Subhashini ChellappanCopyright 2015, WILEY INDIA PVT. LTD.Import data from a CSV fileBig Data and Analytics by Seema Acharya and Subhashini ChellappanCopyright 2015, WILEY INDIA PVT. LTD.Given a CSV file “sample.txt” in the D: drive, import the file into the MongoDBcollection, “SampleJSON”. The collection is in the database “test”.`Mongoimport -db test -collection SampleJSON -type=csv -headerline -file d:.txt-headerline` instructs mongoimport to determine the name of the fields using the first line in the CSV file(If using -type csv or -type tsv)-fieldFile =<filename> option allows you to specify a file that holds a list of field names if your CSV or TSV file does not include field names in the first line of the file.-file=<filename> specifies the location and name of a file containing the data to import. If you do not specify a file, mongoimport reads data from standard input.MongoExportBig Data and Analytics by Seema Acharya and Subhashini ChellappanCopyright 2015, WILEY INDIA PVT. LTD.Export data to a CSV fileBig Data and Analytics by Seema Acharya and Subhashini ChellappanCopyright 2015, WILEY INDIA PVT. LTD.This command used at the command prompt exports MongoDB JSON documents from “Customers” collection in the “test” database into a CSV file “Output.txt” in the D: drive.`-db test -collection Customers -csv -fieldFiled:.txt -outMongoexportd:.txt`Answer a few quick questions ...Big Data and Analytics by Seema Acharya and Subhashini ChellappanCopyright 2015, WILEY INDIA PVT. LTD.CrosswordBig Data and Analytics by Seema Acharya and Subhashini ChellappanCopyright 2015, WILEY INDIA PVT. LTD.Answer MeBig Data and Analytics by Seema Acharya and Subhashini ChellappanCopyright 2015, WILEY INDIA PVT. LTD.□ What is MongoDB?□ Comment on Auto-sharding in MongoDB.□ What are collections and documents?□ What is JSON?□ Explain your understanding of Update In-Place.Summary please...Big Data and Analytics by Seema Acharya and

Subhashini ChellappanCopyright 2015, WILEY INDIA PVT. LTD.Ask a few participants of the learning program to summarize the lecture.References ...Big Data and Analytics by Seema Acharya and Subhashini ChellappanCopyright 2015, WILEY INDIA PVT. LTD.Further ReadingsBig Data and Analytics by Seema Acharya and Subhashini ChellappanCopyright 2015, WILEY INDIA PVT. LTD.<http://www.mongodb.org/> <https://university.mongodb.com/> <http://www.tutorialspoint.com/mongodb/>Thank youBig Data and Analytics by Seema Acharya and Subhashini ChellappanCopyright 2015, WILEY INDIA PVT. LTD.